

Assignment No 3

Association Data Mining

In this Assignment you will explore unsupervised knowledge discovery techniques of association rule mining and sequential patterns mining. In particular you will apply the priori algorithm on a couple of real-world data set of a marketing department of a financial firm analysis.

The marketing department of a financial firm keeps records on customers, including demographic information and the type of accounts. When launching a new product, such as a "Personal Equity Plan" (PEP), a direct mail piece, advertising the product, is sent to existing customers, and a record kept as to whether that customer responded and bought the product. Based on this store of prior experience, the managers decide to use data mining techniques to build customer profile models.

The data is contained in the file **bank-data.csv** which is accessible through the course's Moodle. Each record is a customer description where the "pep" field indicates whether or not that customer bought a PEP after the last mailing.

Your goal is to perform Association Rule discovery on the data set using

- 1. The priori algorithm.**
- 2. Frequent-Pattern growth algorithm (FP growth algorithm)**

Note: Association rule mining requires discretization of continuous variables. First perform the necessary preprocessing steps required for association rule mining. Specifically, the "id" field will need to be removed and the numerical attributes must be discretized.

Next perform association rule discovery on the transformed data. Experiment with different parameters so that you get at least 80-100 strong rules (e.g., rules with high lift and confidence which at the same time have relatively good support). Then:

- List the top 10 popular items for each algorithm
- Bar plot visualization of the 10 top popular items for each algorithm
- scatterplot of the rules was generated i.e. scatter plot between support and confidence of the generated rules for each algorithm
- Select the top 5 most "interesting" rules and for each specify the following from each algorithm:
 - An explanation of the pattern and why you believe it is interesting based on the business objectives of the company;
 - Any recommendations based on the discovered rule that might help the company to better understand behavior of its customers or in its marketing campaign.

Note: The top 5 most interesting rules are most likely not the top 5 in the result set. They are rules that, in addition to having high support, lift, and confidence, also provide some non-trivial, actionable knowledge based on the underlying business objectives.